on her clock. This is no sleight of hand: Gracie will have to search her distant memory, some 60 years before, to recall passing George in space. For George, on the other hand, it was a mere 6 years ago. In a real sense, George's motion has made him a time traveler, albeit in a very precise sense: He has traveled into Gracie's future.

Getting the two clocks back together for direct comparison might seem to be merely a logistical nuisance, but it is really at the heart of the matter. We can imagine a variety of tricks to circumvent this chink in the paradox armor, but all ultimately fail. For instance, rather than bringing the clocks back together, what if George and Gracie compare their clocks by cellular telephone communication? If such communication were instantaneous, we would be faced with an insurmountable inconsistency: reasoning from Gracie's perspective, George's clock is running slow and hence he must communicate less elapsed time; reasoning from George's perspective, Gracie's clock is running slow and hence she must communicate less elapsed time. They both can't be right, and we would be sunk. The key point of course is that cell phones, like all forms of communication, do not transmit their signals instantaneously. Cell phones operate with radio waves, a form of light, and the signal they transmit therefore travels at light speed. This means that it takes time for the signals to be received—just enough time delay, in fact, to make each perspective compatible with the other.

Let's see this, first, from George's perspective. Imagine that every hour, on the hour, George recites into his cell phone, "It's twelve o'clock and all is well," "It's one o'clock and all is well," and so forth. Since from his perspective Gracie's clock runs slow, at first blush he thinks that Gracie will receive these messages prior to her clock's reaching the appointed hour. In this way, he concludes, Gracie will have to agree that hers is the slow clock. But then he rethinks it: "Since Gracie is receding from me, the signal I send to her by cell phone must travel ever longer distances to reach her. Maybe this additional travel time compensates for the slowness of her clock." George's realization that there are competing effects—the slowness of Gracie's clock vs. the travel time of his signal—inspires him to sit down and quantitatively work out their combined effect. The result he finds is that the travel time effect more than compensates for the slowness of